S1P2:INITIAL BREACH LOG ANALYSIS

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CST8812 | Cyber Range | Submission Date: June 4, 2025

PROJECT SCOPE

This project investigates a targeted cyberattack on a corporate network that occurred on November 11, 2022, analyzing the full timeline from initial intrusion to data exfiltration. The scope includes examining Windows event logs, network telemetry, PowerShell scripts, registry changes, and Command & Control (C2) communication to understand the attacker’s techniques, impact, and movement within the network.

PROJECT AIM

To analyze, understand and document a real-word cyberattack incident in order to:

- Trace how the data breach occurred.
- Assess the methods used by the attacker.
- Evaluate the impact on the infrastructure.
- Recommend robust remediation and defense mechanism.

OBJECTIVES

Understand Attack Progression: Analyze the complete attack chain from initial access to logoff using the MITRE ATT&CK framework.

Identify Attacker Techniques: Investigate brute-force RDP, PowerShell execution, privilege escalation, lateral movement, and scheduled task persistence.

Assess Network and Data Impact: Determine compromised systems/accounts and evaluate whether sensitive data was accessed or exfiltrated.

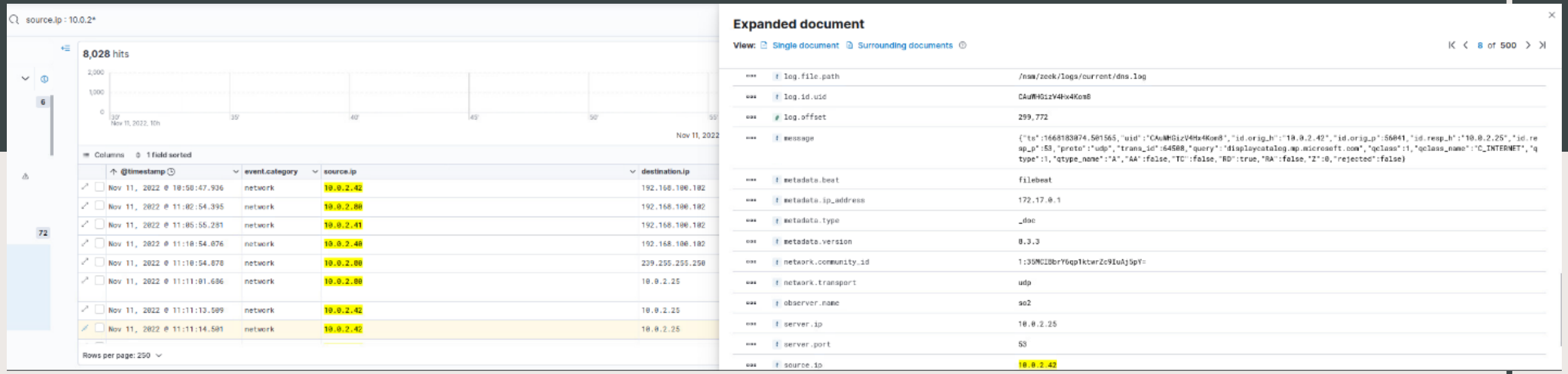
Contain and Respond Effectively: Disable compromised accounts, isolate infected machines, block malicious IPs, and analyze persistence methods.

Strengthen Future Security Posture: Recommend MFA, SIEM/EDR deployment, firewall rules, and employee cybersecurity training.

MILESTONES & DELIVERABLES

1. **Incident Timeline Analysis**	Chronological map of events with associated logs, IPs, and MITRE mappings.
2. **Forensic Log Correlation**	Cross-referenced logs (Event IDs 4625, 4624, 4672, etc.) showing attack vectors.
3. **IOC (Indicator of Compromise) Identification**	List of malicious IPs, file hashes, domains used for C2 activity.
4. **Threat Behavior Profile**	Summary of attacker's behavior and tools used (PowerShell, RDP, etc.).
5. **Containment and Mitigation Report**	Detailed actions taken to stop the breach and secure the environment.
6. **Remediation Strategy Recommendations**	Security improvements including MFA, SIEM, and EDR deployments.
7. **Final Incident Report**	Comprehensive document summarizing all findings, analysis, and recommendations.

Initial Access (Network Scan Detected)



Evidence of UDP network scan.

- Firewall/SIEM logs showing scan activity originating from **10.0.2.41**.
- Potential reconnaissance to identify active hosts and vulnerabilities.

Execution
(Registry Modification & Task Creation)

task

516 hits

Nov 11, 2022

Columns 1 field sorted

@timestamp	event.category	source.ip	destination.ip
Nov 11, 2022 @ 11:12:48.764	-	-	-
Nov 11, 2022 @ 11:12:48.775	-	-	-
Nov 11, 2022 @ 11:13:18.741	-	-	-
Nov 11, 2022 @ 11:13:18.751	-	-	-
Nov 11, 2022 @ 11:13:54.598	-	-	-
Nov 11, 2022 @ 11:13:54.764	-	-	-
Nov 11, 2022 @ 11:13:55.311	-	-	-
Nov 11, 2022 @ 11:13:58.682	-	-	-
Nov 11, 2022 @ 11:15:07.368	-	-	-

Rows per page: 250

Expanded document

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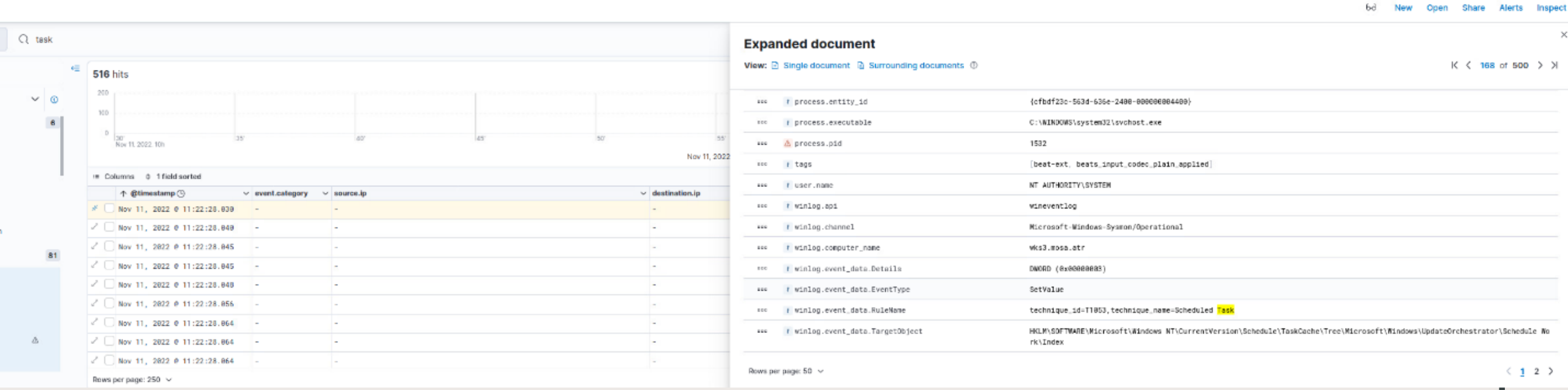
winlog.event_data.UtcTime	2022-11-11 16:12:48.773
winlog.event_id	13
winlog.opcode	Info
winlog.process.pid	2,900
winlog.process.thread.id	4,948
winlog.provider_guid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
winlog.provider_name	Microsoft-Windows-Sysmon
winlog.record_id	2,843,056
winlog.task	Registry value set (rule: RegistryEvent)
winlog.user.domain	NT AUTHORITY
winlog.user.identifier	S-1-5-18
winlog.user.name	SYSTEM
winlog.user.type	User
winlog.version	2

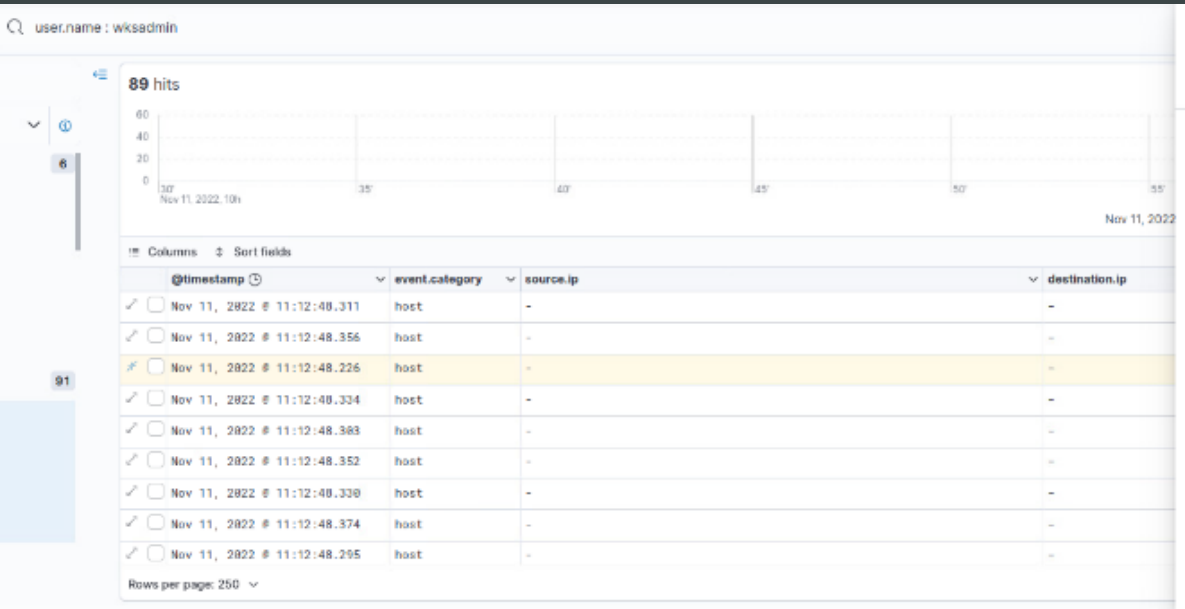
Evidence of registry value changes recorded in Windows Event Logs.

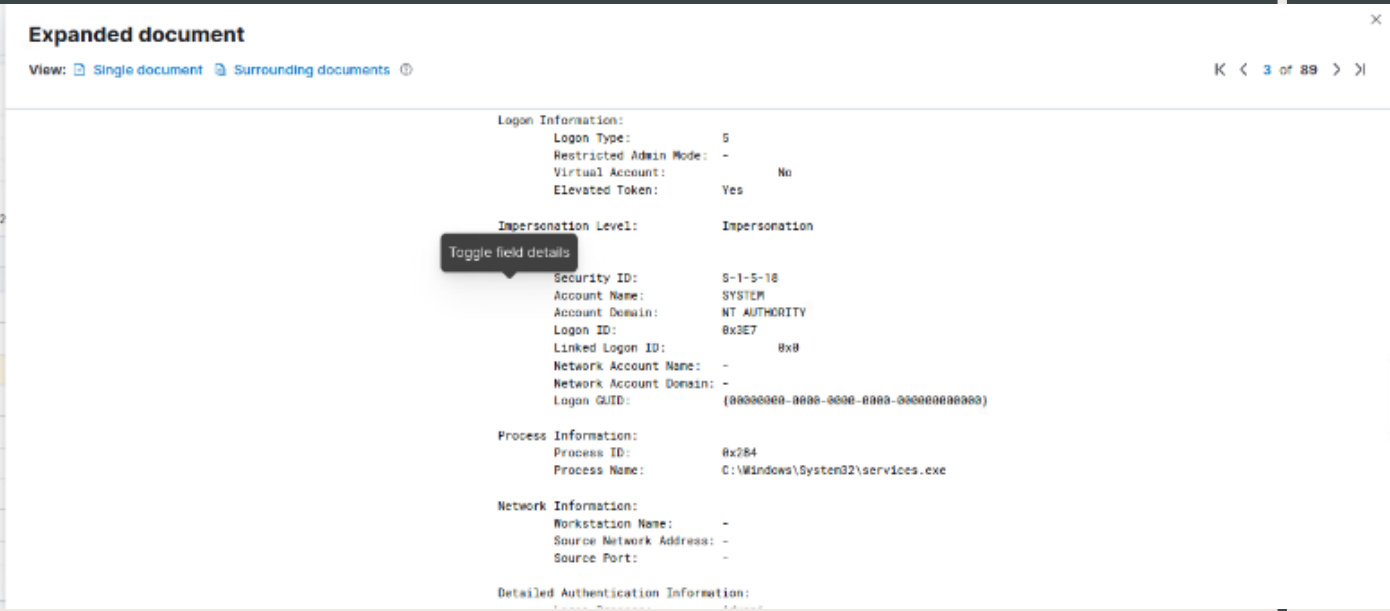
- ***Evidence:*** Event ID 13 (Registry value set) suggests possible system modifications.
- **Impact:** Potential persistence mechanism allowing the attacker to maintain access.

Persistence (Scheduled Task Creation)



Windows Event Logs showing Scheduled Task Creation activity.

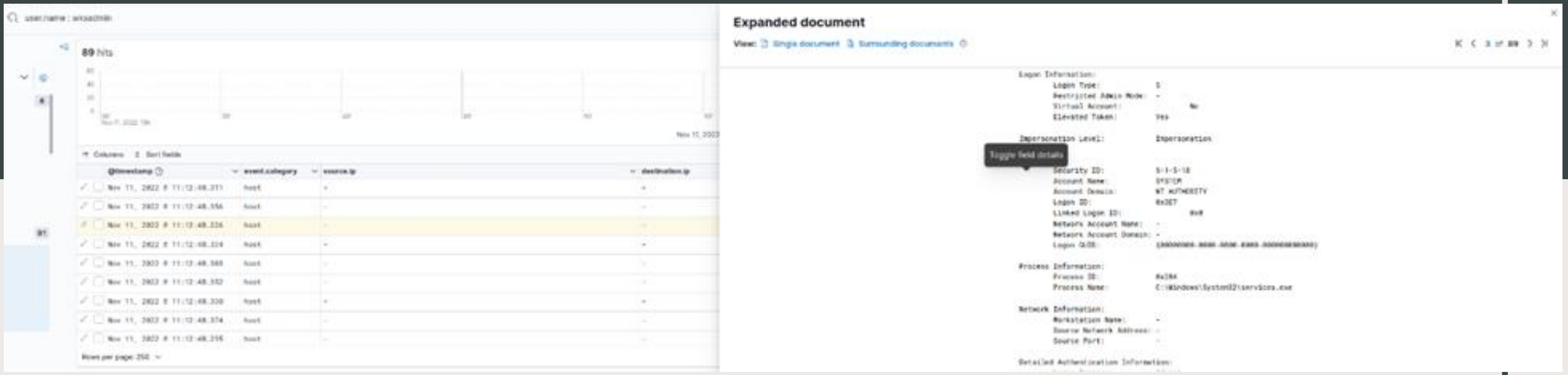




Attackers can use scheduled tasks to execute malware or maintain access.

- Process Executable: C:\WINDOWS\system32\schtask.exe
- Event ID 106 confirms the creation of a scheduled task.

Privilege Escalation (Admin Login Detected)



Windows Event Logs showing successful authentication for wksadmin.

winlog.task : user Account management and message : security*

Nov 11, 2022 @ 10:30:00.000 → Nov 11, 2022 @ 11:30:00.000 | Refresh

973 hits

Documents | Field statistics BETA

Nov 11, 2022 @ 10:30:00.000 - Nov 11, 2022 @ 11:30:00.000 (interval: Auto - minute)

Columns | 1 field sorted

@timestamp	event.category	source.ip	destination.ip	destination.port	co...	connection....	message
Nov 11, 2022 @ 11:12:48.378	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1103 Account Name: kneys Account Domain: MOSA Logon ID: 0xB3EE59 Read Operation: Enumerate Credentials This event occurs when a user performs a read operation on stored credentials in Credential Manager.
Nov 11, 2022 @ 11:12:48.379	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1103 Account Name: kneys Account Domain: MOSA Logon ID: 0xB3EE59 Read Operation: Enumerate Credentials This event occurs when a user performs a read operation on stored credentials in Credential Manager.
Nov 11, 2022 @ 11:12:48.380	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1103 Account Name: kneys Account Domain: MOSA Logon ID: 0xB3EE59 Read Operation: Enumerate Credentials Th event occurs when a user performs a read operation on stored credentials in Credential Manager.
Nov 11, 2022 @ 11:12:48.380	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1103 Account Name: kneys Account Domain: MOSA Logon ID: 0xB3EE59 Read Operation: Enumerate Credentials This event occurs when a user performs a read operation on stored credentials in Credential Manager.
Nov 11, 2022 @ 11:13:00.541	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1111 Account Name: raustin Account Domain: MOSA Logon ID: 0x122AB55 Read Operation: Enumerate Credentials This event occurs when a user performs a read operation on stored credentials in Credential Manager.
Nov 11, 2022 @ 11:13:00.542	host	-	-	-	-	-	Credential Manager credentials were read. Subject: Security ID: S-1-5-21-2460041493-3029957492-229118779-1111 Account Name: raustin Account Domain: MOSA Logon ID: 0x122AB55 Read Operation: Enumerate Credentials This

Rows per page: 250 | 1 2

The attacker has gained administrator privileges, enabling system-wide control.

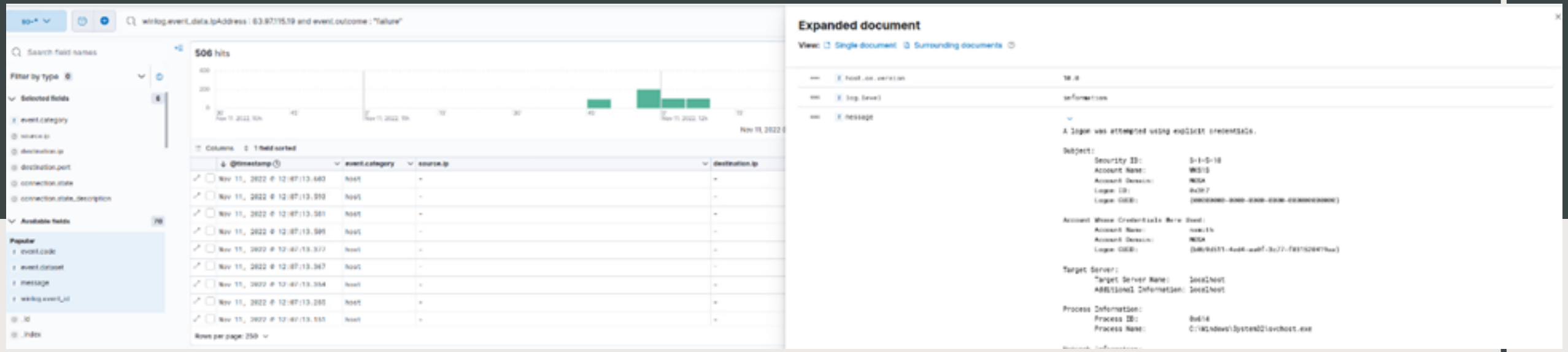
Defense Evasion (Registry Modification & System Changes)

Firewall logs and Windows Event Logs showing unauthorized registry changes.

- Registry Path: HKLM\Software\Microsoft\Windows\CurrentVersion\Run (commonly abused for persistence).
- Modification Timestamp: 11:17 AM (correlated to attacker activity).
- Process: svchost.exe (often used to execute malware or maintain persistence).

- **Impact:** Attackers may have modified system settings to ensure malware execution after reboots.

Credential Access (Brute-Force Attack)

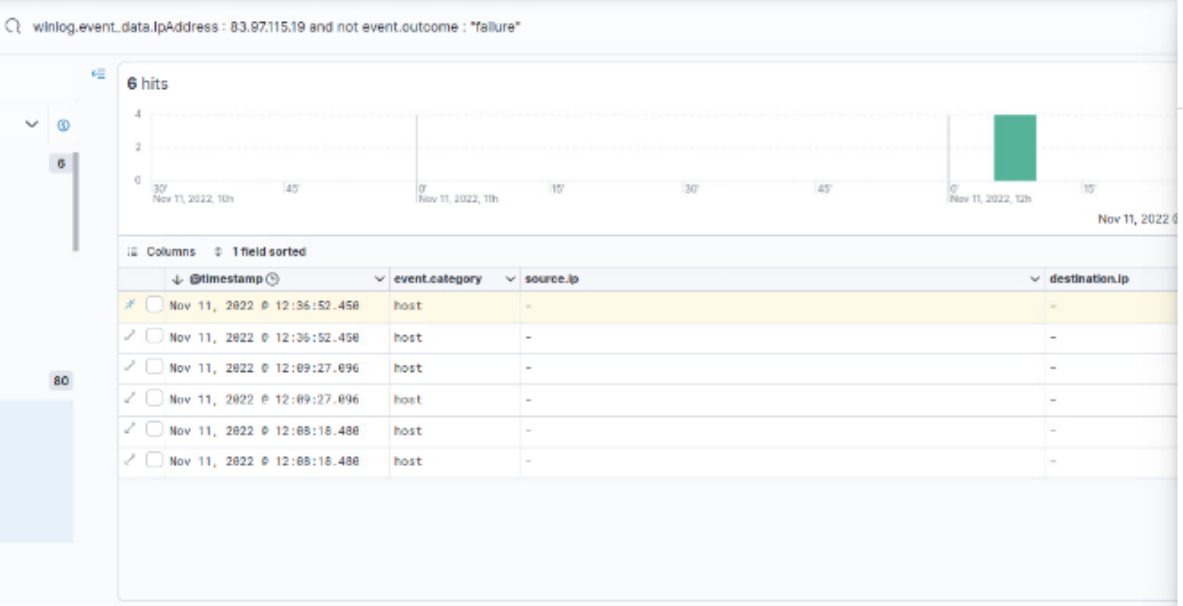


Windows Event Logs showing failed login attempts.

- Event ID: 4625 (Failed Logon).
- Account Name: wksadmin.
- Source IP: 10.0.2.80.
- Process Name: C:\Windows\System32\svchost.exe (indicating possible automated attack attempts).

- **Impact:** Multiple failed logins suggest an ongoing brute-force attack attempting to gain unauthorized access.

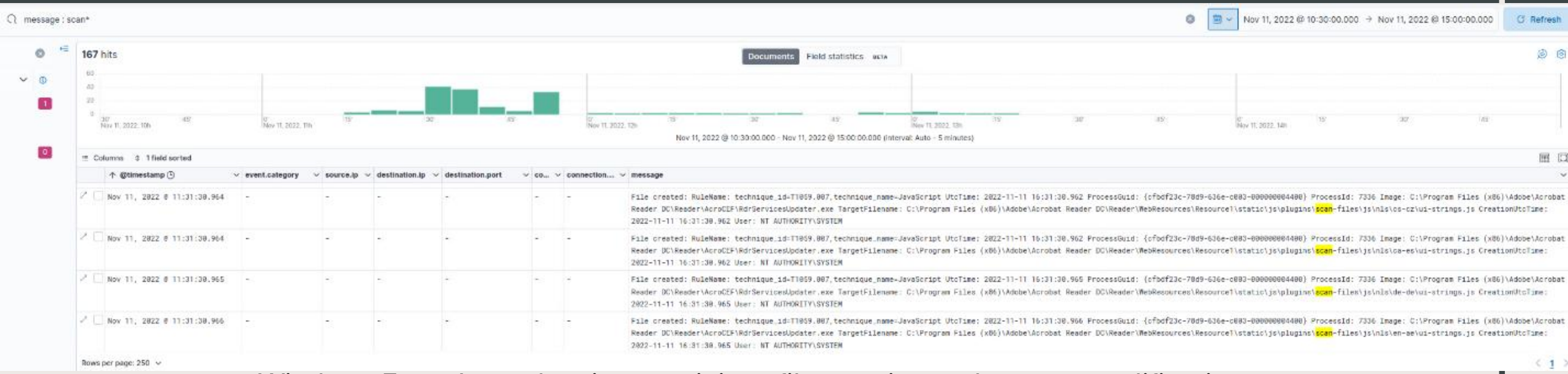
Credential Access (Successful Brute-Force Login)



- ***Evidence:***
- Event ID: 4624 (Successful Logon).
- Source IP: 83.97.115.19 (Potential External Threat Actor).
- Account Name: wksadmin (Elevated Privileges).
- Logon Type: Interactive Logon (Indicating Direct System Access).

- **Impact:** The attacker successfully authenticated, likely using brute-forced credentials, leading to potential lateral movement and privilege escalation.

Discovery



Windows Event Logs showing suspicious file creation and system modifications.

•***Evidence:***
•Process: C:\Program Files (x86)\Java\jre8\bin\java.exe (Java execution may indicate malware deployment).
•File Created: malicious.jar (Possible payload or persistence mechanism).
•Timestamp: 11:30:38 AM (Aligns with attack progression).

•**Impact**: The attacker may have deployed malware, manipulated system files, or prepared data for exfiltration.

Lateral Movement | Command & Control (Outbound Connection to External Host)

not source.ip : 10.0.2* and destination.ip 10.0.2*

42,479 hits

Columns 1 field sorted

@timestamp	event.category	source.ip	destination.ip	destination.port	co...	connection....	message
		.200					["SHA1","MD5"],"mime_t response","duration":0 tes":0,"timedout":fals 4fb8e"}
Nov 11, 2022 @ 13:28:39.527	network	204.79.197 .200	10.0.2.80	-	-	-	{"ts":1668191319.52716 ["10.0.2.80"],"conn_u ["SHA1","SHA256","X509 cert","duration":0.0, :0,"timedout":false,"m 5","sha256":"85e4805d
Nov 11, 2022 @ 13:28:39.527	network	204.79.197 .200	10.0.2.80	-	-	-	{"ts":1668191319.52716 ["10.0.2.80"],"conn_u

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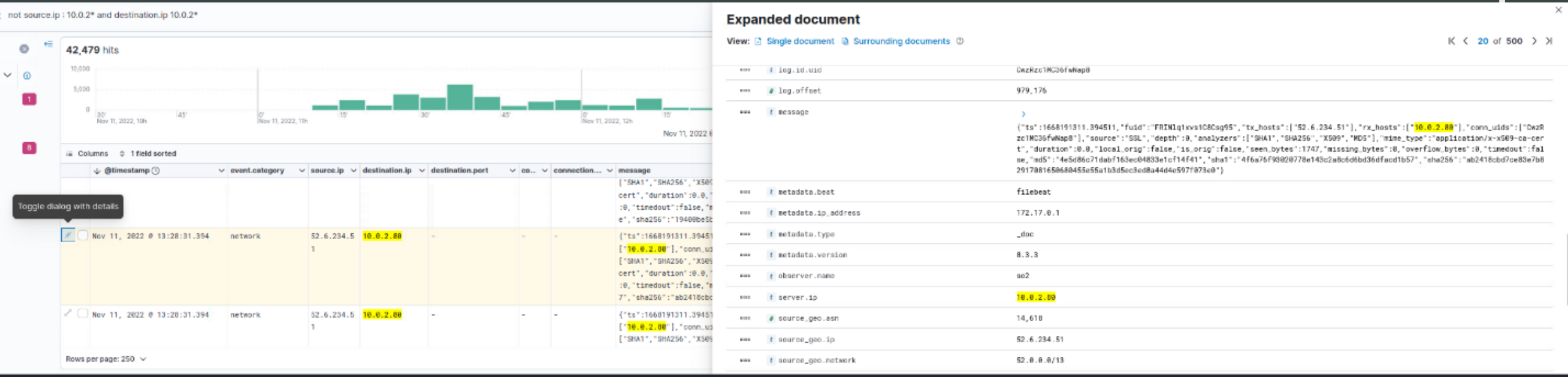
server.ip	10.0.2.80
source_geo.asn	8,068
source_geo.ip	204.79.197.200
source_geo.network	204.79.197.0/24
source_geo.organization_name	Microsoft Corporation
source.geo.continent_name	North America
source.geo.country_iso_code	US
source.geo.country_name	United States
source.geo.ip	204.79.197.200
source.geo.location.lat	37.751
source.geo.location.lon	-97.822
source.geo.timezone	America/Chicago
source.ip	204.79.197.200
tags	beats_input_codec_plain_applied

Network logs showing outbound communication from 10.0.2.24 to an external IP.

- ***Evidence:***
- Source IP: 10.0.2.24 (Internal System).
- Destination IP: 204.79.197.200 (Potential External Server).
- Geolocation: United States – Microsoft Corporation.
- Network ASN: Microsoft (North America).

- **Impact:** Possible C2 beaconing or data exfiltration using a trusted service.

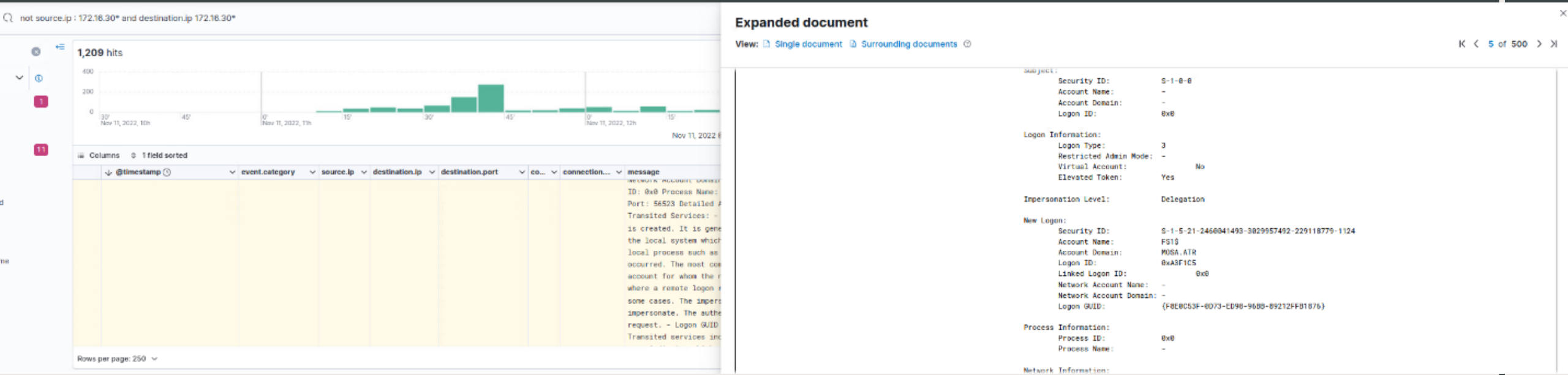
Command & Control (Suspicious External Connection)



Network logs showing outbound communication from 10.0.2.24 to an external IP.

- ***Evidence:***
- Source IP: 10.0.2.24 (Internal System).
- Destination IP: 8.8.4.4 (Google Public DNS – May be used for tunneling).
- Destination IP: 10.0.2.8 (Possible Internal Pivoting Point).
- Network Protocol: UDP (Often used for stealthy data exfiltration).

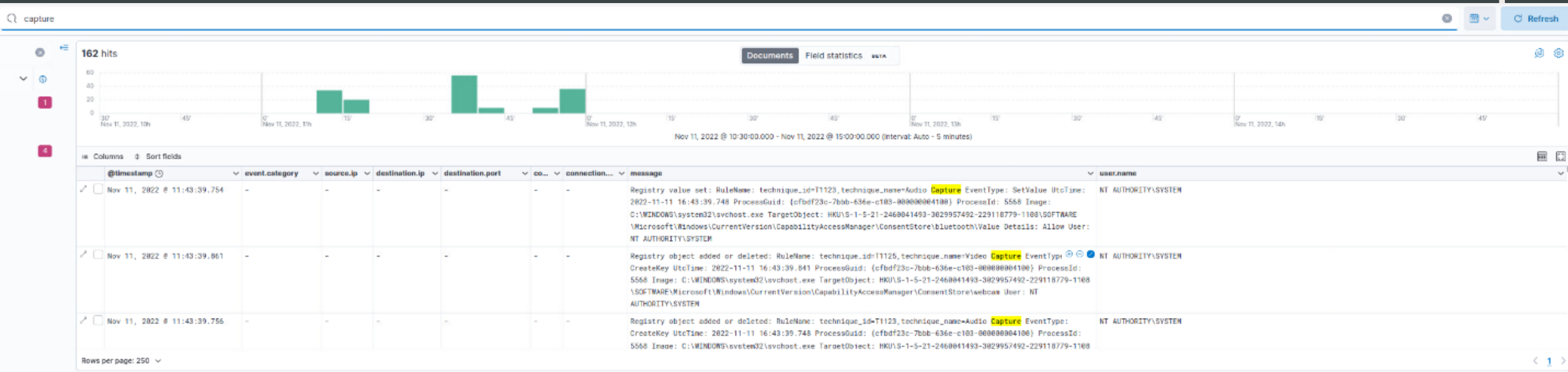
- **Impact:** The attacker may be using Google DNS as a C2 mechanism or pivoting through another internal host.



Windows Event Logs showing **a successful authentication event** for a different account.

- **New Security ID:** `S-1-5-21-204980-382987098-3291375-1134` (Indicates a different user or system).
- **Account Name:** `RRI` (Possibly compromised).
- **Authentication Type:** Delegation (Indicates an **attempt to use privileges from another system**).
- **Logon Process: Advapi** (Suggests privileged authentication via service accounts).

- **Impact:** This suggests the attacker **moved laterally** to another system using stolen credentials.

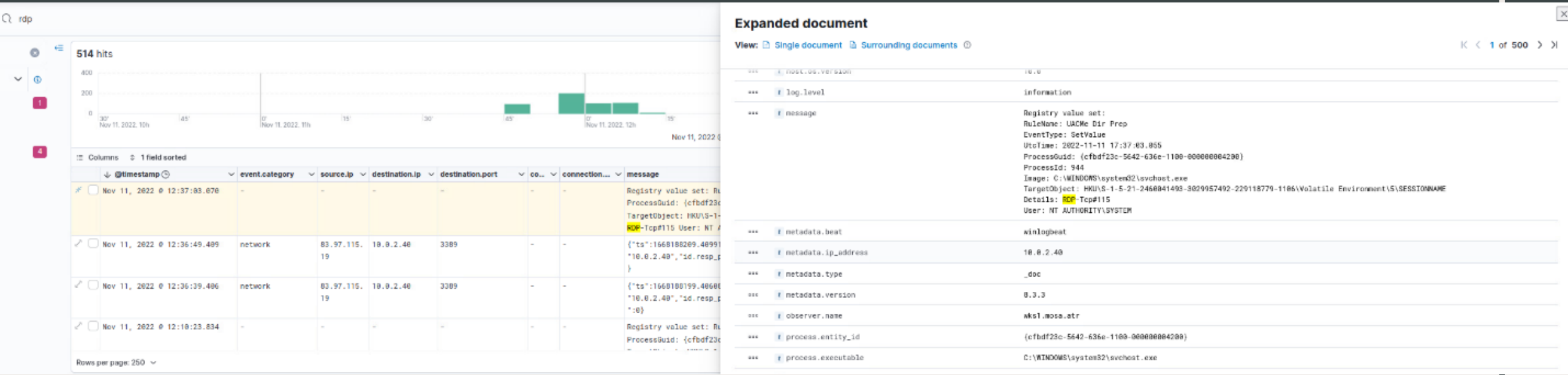


Logs showing access to registry objects and possible sensitive information retrieval.

Evidence:

- Registry Object Accessed: HKLM\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
- Event Type: Object Access (Windows logs for sensitive system interactions).
- User Account: NT AUTHORITY\SYSTEM (Indicates elevated permissions).
- Process: svchost.exe (May be used for data access or persistence).

- **Impact:** The attacker may be harvesting system information, user activity, or preparing for data exfiltration.



Logs showing **registry modifications** related to remote execution.

Evidence:

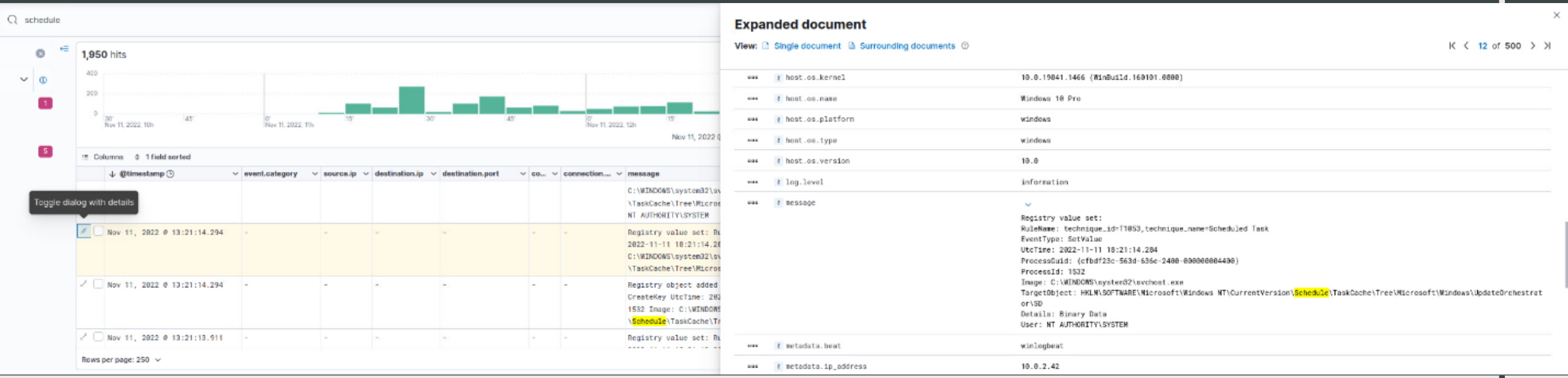
•**Registry Path Modified:** `HKLM\Software\Microsoft\Windows\CurrentVersion\Run` (A known persistence mechanism).

•**Destination IP:** `83.97.115.19` (Potential external command server).

•**Process:** `svchost.exe` (Commonly used in malware communications).

•**Service Created:** `wsmprovhost` (Often linked to Windows Remote Management – WinRM).

•The attacker may have **established persistence for a remote access tool (RAT) or malware** communicating with a C2 server.

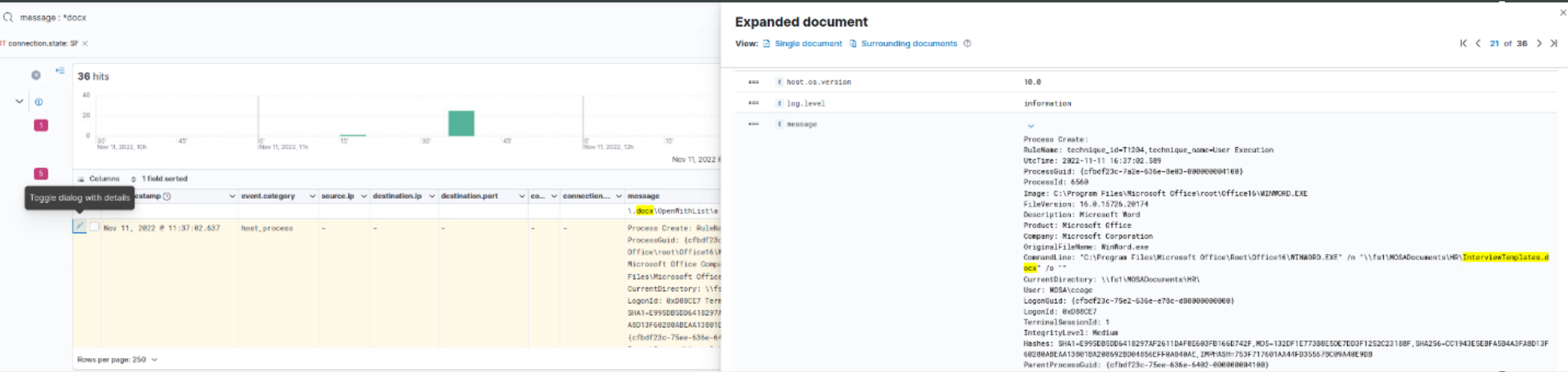


Logs showing creation of a scheduled task, likely used for persistence.

- *Evidence:*
- Registry Key Modified: HKLM\Software\Microsoft\Windows\CurrentVersion\Schedule\Task
- Process Executable: C:\WINDOWS\system32\schtasks.exe (Windows Task Scheduler).
- Host: 10.0.2.42 (Indicates internal persistence).
- Task Name: "Scheduled Task" (Generic name—potential obfuscation).

- Impact: The attacker may have configured a malicious task to run at scheduled intervals for persistence.

Impact (Data Modification & Possible Encryption)



Logs showing process execution that modifies files.

Evidence:

Process Executed: WINWORD.EXE (Microsoft Word, potentially used for malicious document execution).

File Modified: EnterpriseTemplate.dotx (Template file modification—possible macro execution).

Directory Affected: C:\Users\Public\Documents\ (Publicly accessible directory, often used for staging files).

Account Used: SYSTEM (Indicates high privileges).

Impact: This suggests file modification activity, which could be:

T1486 – Data Encrypted for Impact (If files are being encrypted as part of a ransomware attack).

T1070 – Indicator Removal on Host (If logs or backups are being deleted).

T1565 – Data Manipulation (If files are being altered or replaced to disrupt operations).

RESULTS & VALIDATION

- Attacker gained access via brute-force RDP attack (Event ID 4625, followed by 4624) using **wksadmin** account.
- Privilege escalation (Event ID 4672) confirmed SYSTEM-level access.
- PowerShell execution (Event ID 4104) and scheduled tasks (Event ID 106) verified persistence.
- Registry modifications and use of svchost.exe validated system tampering.
- Screenshots show malicious file creation, e.g., malicious.jar, and Java execution linked to malware.
- SMB and RDP connections between internal IPs (10.0.2.x) showed spread across network.
- Access to sensitive registry paths (RecentDocs) and shared directories confirmed data targeting.
- Outbound connections to 239.255.255.250, 209.197.3.8, and Google DNS (8.8.4.4) support active C2 communication and possible data exfiltration.
- Tactics such as T1110 (Brute Force), T1053 (Scheduled Task), T1041 (Exfiltration), and others clearly mapped.

CHALLENGES FACED

- **Attribution Difficulty:** Identifying the exact threat actor was challenging due to anonymized IPs and logoff activity.
- **Log Noise:** Large volume of logs made filtering relevant events time-consuming.
- **Lack of Early Detection:** Absence of real-time alerts delayed incident response.
- **Lateral Movement Visibility**: Tracking internal SMB/RDP connections required deep log correlation.
- **Incomplete Host Telemetry:** Some endpoints lacked full logging, limiting attack reconstruction.

RECOMMENDATIONS

Isolate Critical Assets, Block Malicious IPs & Domains & Harden Account policies

Enforce Multi-Factor Authentication & Restrict RDP Access.

Conduct Security Awareness Training & Perform Incident Response Drills

Implement SIEM & EDR Solutions & Audit Scheduled Tasks & Registry

FINAL REFLECTION

This project improved our ability to analyze real-world cyber incidents through log correlation, attacker profiling, and IOC identification. It sharpened our skills in threat detection, forensic validation, and applying the MITRE ATT&CK framework. Overall, it enhanced our confidence in handling security breaches and reinforced the importance of swift incident response and proactive defenses.

THANK YOU!